



IndianOil

JOURNEY RISK MANAGEMENT (JRM) STUDY

Salem Terminal TO N. S. BHARANI FUELS

Objective of the JRM Report

This JRM report is designed to ensure compliance with the Central Motor Vehicle Rules, 1989 (CMVR), AIS 140 standards, and the Road Transport Safety Policy (RTSP). It provides a comprehensive risk assessment for the transportation of hazardous materials along specified routes. By integrating these legal frameworks, the report offers a broad strategy for identifying and mitigating route-specific risks.

Regulatory Compliance

The report complies with the Central Motor Vehicles (Eleventh Amendment) Rules, 2022, mandating safe transportation practices for N2 and N3 category vehicles carrying hazardous materials. These rules require detailed route assessments, especially regarding road conditions, speed limits, and risk areas, to ensure safety compliance.

Risk Management Strategy

This report categorizes transportation routes into high-risk and medium-risk areas, with a focus on factors such as sharp turns, accident-prone regions, and elevation changes. The goal is to provide actionable

recommendations to minimize these risks, including speed regulations, driver warnings for hazardous zones, and the option of alternate routes.

Compliance with the Road Transport Safety Policy (RTSP)

The report integrates RTSP provisions, including mandatory driving hours, rest periods, and nighttime driving restrictions. It ensures that drivers follow official guidelines, such as taking prescribed rest breaks and avoiding dangerous road conditions like poor visibility, heavy crowds, or high-traffic areas during peak hours.

Emergency Preparedness and Response

The report highlights the significance of predetermined emergency stops for refueling, rest, and overnight stays. It includes protocols for safe responses to road hazards, alternative routes, and rerouting processes if roads are closed or severe weather arises. This aligns with the RTSP emphasis on driver safety and rapid emergency response.

Environmental Considerations

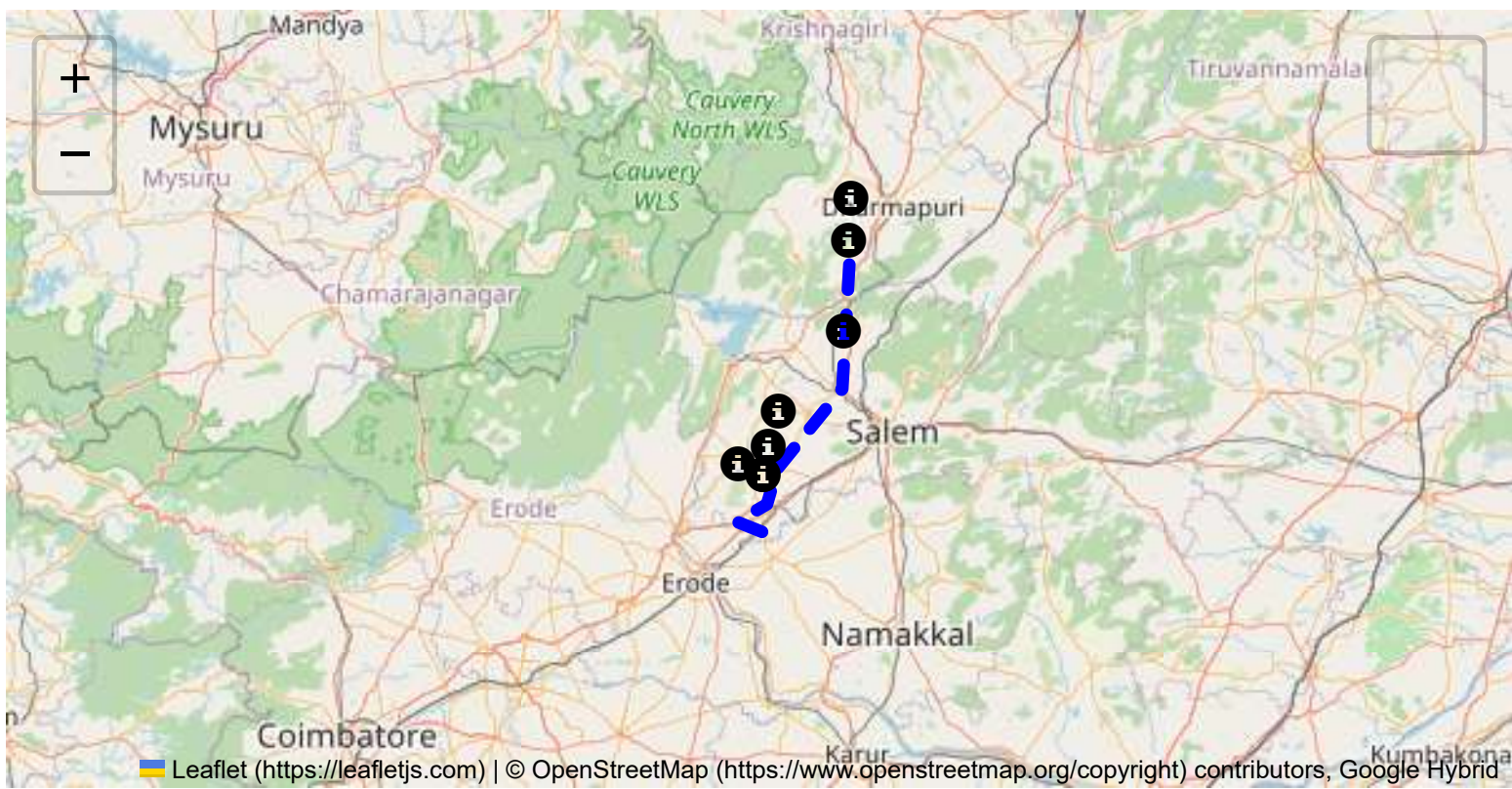
The JRM report addresses environmental risks along the route, ensuring compliance with environmental protection laws in ecologically sensitive zones. It suggests strategies such as identifying areas near water bodies, forests, or populated regions and implementing safety measures to minimize environmental impacts during transport.

Journey Risk Mitigation

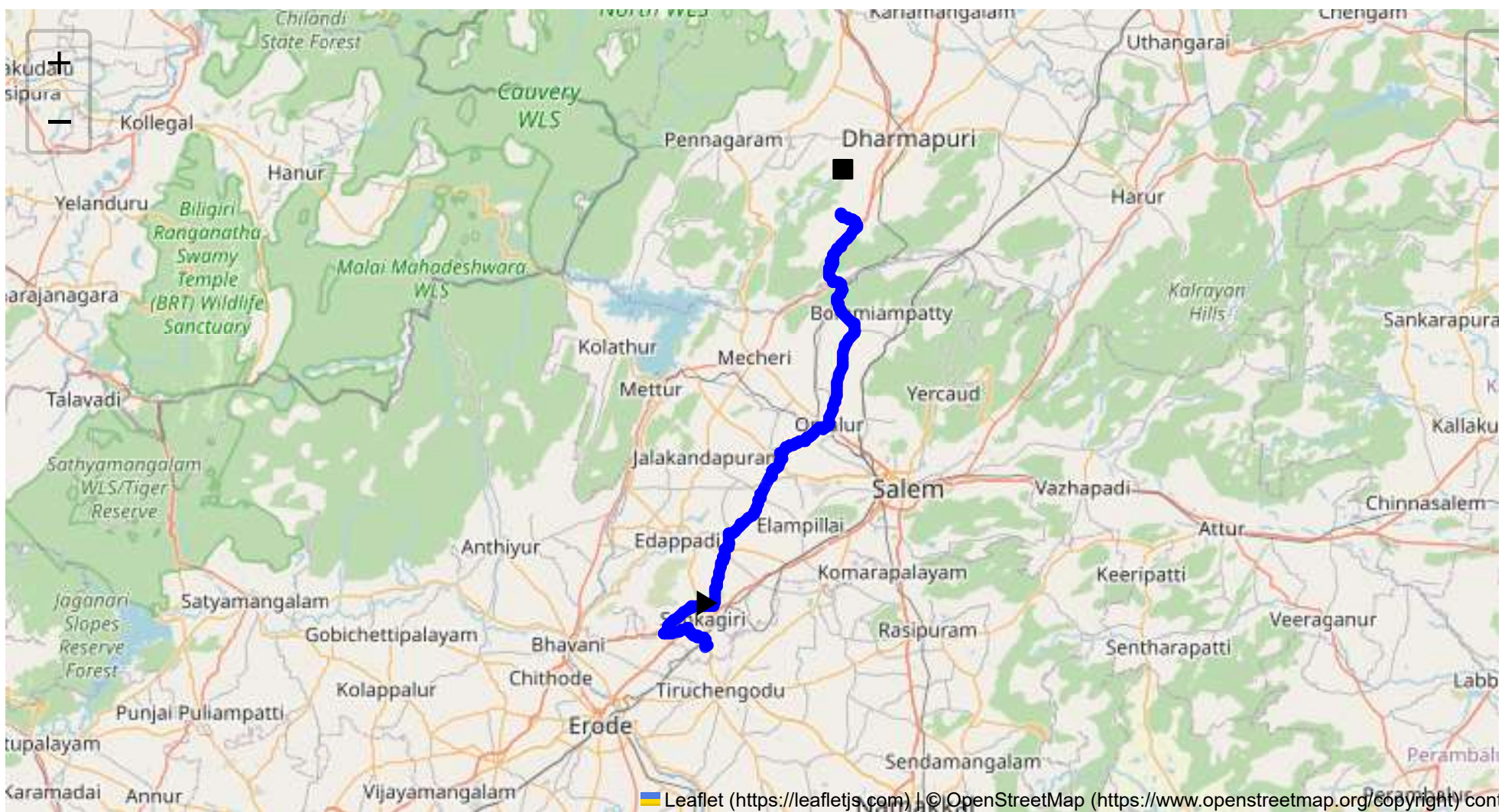
The report includes route-specific risk assessments, detailed journey charts, and defensive driving guidelines for each transport route. Integration with vehicle tracking systems guarantees real-time warnings on hazardous areas, speed limits, and mandatory stops, consistent with RTSP and CMVR safety norms.

Compliance with Government Directives

This report fully adheres to governmental directives regarding hazardous material transportation, implementing mandatory speed limits, nighttime driving restrictions, and comprehensive driver briefings and real-time alerts about route-related risks.



Route Summary:
Total Distance: 92.71 km
Estimated Duration: 2.1 hours
Adjusted Duration (Heavy Vehicle): 2.6 hours
Start: (11.4381, 77.8734)
End: (12.0303464, 78.0630407)



Welcome to the Journey Risk Management Study

1. Overview of the Route Map

The route begins at Sangagiri and winds through several key points, eventually reaching Samichettipatti. The journey spans approximately 92.71 kilometers. The route traverses towns like Puthur, Padaiveedu, Konganapuram, Omalur, and Thoppur. The major highways involved include the Bhavani Main Road (NH 544H) and Salem - Kochi Highway, along with State Highway 86.

2. Typical Weather Conditions and Potential Weather-Related Hazards

The weather in Tamil Nadu can be hot and humid, particularly from March to May. The monsoon season, spanning June to September, can bring heavy rains, potentially causing flooding and reducing visibility. The dry season (October to March) generally offers more stable driving conditions but be wary of sudden thunderstorms.

3. Analysis of Traffic Patterns

Traffic congestion is common in towns like Omalur and along major highways. The peak hours are typically from 8:00 AM to 10:00 AM and from 5:00 PM to 7:00 PM, where major congestion can occur. Markets and schools can cause additional local slowdowns. Attention should be paid to intersections and toll booths as they can cause delays.

4. Assessment of Road Quality and Infrastructure

While the main highways like the Salem - Kochi Hwy and SH 86 are generally well-maintained, some connecting roads may have potholes or uneven surfaces. Signage is decent on main roads but might be lacking on rural or less-traveled paths.

5. Suggestions for Alternative Routes for Emergencies

In the case of major obstructions or hazards, consider using NH 544 for an alternate parallel route eastwards. The local roads connecting nearby towns also offer alternate connectivity but should be approached with caution for heavy vehicles due to their varying quality.

6. Summary of Local Regulations Affecting Hazardous Material Transport

Transport of hazardous materials must comply with the regulations set by the Government of Tamil Nadu. Standard protocols include compliance with hazardous warning placards, proper documentation, and ensuring the presence of emergency response equipment.

7. Overview of Historical Incidents Involving Heavy Vehicles or Hazardous Materials

There have been incidents involving heavy vehicles where road conditions or human error were factors. Notably, accidents involving chemical spills have heightened awareness around hazardous transport on these routes, emphasizing the need for driver vigilance and proper training.

8. Environmental Considerations and Sensitive Areas

The route passes through agricultural land and near water bodies, necessitating extra caution with hazardous materials to prevent contamination. Drivers should exercise caution near fields and adhere to speed limits to minimize dust and noise pollution.

9. Analysis of Communication Coverage

Major towns and highways generally have good mobile network coverage. However, rural areas and less traveled routes might experience dead zones. Backup communication devices or route mapping tools are advised.

10. Estimated Emergency Response Times for Different Route Segments

- Urban Areas (Sangagiri, Omalur): Approximately 15-30 minutes
- Rural Segments (Between towns): Approximately 30-60 minutes
- Remote Areas: Could exceed 60 minutes

11. Overall Summary of Risk Assessment

The route poses a moderate risk for drivers of heavy vehicles carrying hazardous materials. Well-maintained highways offer a generally safe passage, but congestion and varying road conditions in towns and rural areas require caution. Adverse weather conditions during the monsoon season amplify these risks. Continuous driver alertness, adherence to regulations, and preparedness for communication challenges are critical to ensuring a safe journey. Regular reviews of route conditions and emergency protocols should be undertaken to minimize potential hazards.

Risk Assessment - Turns

	Risk Type	Risk Level	Coordinates	Speed Limit
0	Turn	High	11.43961, 77.87341	15 KM/Hr
1	Turn	High	11.44024, 77.87528	15 KM/Hr
2	Turn	High	11.44898, 77.87410	15 KM/Hr
3	Turn	Medium	11.45357, 77.85789	30 KM/Hr
4	Turn	High	11.45352, 77.85698	15 KM/Hr
5	Turn	Medium	11.45847, 77.85140	30 KM/Hr
6	Turn	Medium	11.45898, 77.85135	30 KM/Hr
7	Turn	High	11.46303, 77.84945	15 KM/Hr
8	Turn	Medium	11.45543, 77.81740	30 KM/Hr
9	Turn	High	11.45631, 77.81668	15 KM/Hr
10	Turn	High	11.49383, 77.88442	15 KM/Hr
11	Turn	Medium	11.56083, 77.89921	30 KM/Hr

	Risk Type	Risk Level	Coordinates	Speed Limit
12	Turn	Medium	11.56180, 77.90019	30 KM/Hr
13	Turn	Medium	11.57048, 77.90224	30 KM/Hr
14	Turn	Medium	11.58254, 77.90656	30 KM/Hr
15	Turn	Medium	11.64217, 77.95200	30 KM/Hr
16	Turn	Medium	11.64712, 77.95118	30 KM/Hr
17	Turn	High	11.67858, 77.96834	15 KM/Hr
18	Turn	Medium	11.70062, 77.97961	30 KM/Hr
19	Turn	High	11.70218, 77.97866	15 KM/Hr
20	Turn	Medium	11.71917, 78.01123	30 KM/Hr
21	Turn	Medium	11.72565, 78.01770	30 KM/Hr
22	Turn	Medium	11.73769, 78.03451	30 KM/Hr
23	Turn	Medium	11.73785, 78.03551	30 KM/Hr
24	Turn	High	11.73651, 78.03871	15 KM/Hr
25	Turn	Medium	11.74265, 78.04710	30 KM/Hr
26	Turn	Medium	11.78005, 78.05788	30 KM/Hr
27	Turn	Medium	11.86928, 78.08384	30 KM/Hr
28	Turn	Medium	11.88009, 78.08164	30 KM/Hr
29	Turn	Medium	11.91277, 78.05853	30 KM/Hr
30	Turn	Medium	11.93746, 78.05331	30 KM/Hr
31	Turn	Medium	11.95803, 78.04964	30 KM/Hr
32	Turn	Medium	11.95854, 78.04990	30 KM/Hr
33	Turn	Medium	11.96015, 78.04945	30 KM/Hr
34	Turn	Medium	11.96370, 78.05142	30 KM/Hr
35	Turn	Medium	11.96433, 78.05113	30 KM/Hr
36	Turn	High	11.96542, 78.04938	15 KM/Hr
37	Turn	High	11.96650, 78.04927	15 KM/Hr
38	Turn	Medium	11.96804, 78.05080	30 KM/Hr
39	Turn	High	12.01399, 78.08651	15 KM/Hr
40	Turn	Medium	12.01914, 78.08227	30 KM/Hr
41	Turn	High	12.02037, 78.08260	15 KM/Hr
42	Turn	Medium	12.02084, 78.08193	30 KM/Hr
43	Turn	Medium	12.02139, 78.08167	30 KM/Hr
44	Blind Spot	Blind Spot	12.02306, 78.07557	10 KM/Hr
45	Turn	Medium	12.02328, 78.07564	30 KM/Hr

	Risk Type	Risk Level	Coordinates	Speed Limit
46	Turn	Medium	12.02349, 78.07557	30 KM/Hr
47	Turn	High	12.02465, 78.07231	15 KM/Hr
48	Turn	High	12.02566, 78.07211	15 KM/Hr
49	Turn	Medium	12.02707, 78.06514	30 KM/Hr
50	Turn	Medium	12.03020, 78.06332	30 KM/Hr

Emergency Locations

	type	name	coordinates	speed_limit	risk_level
1	hospital	Sri Shanthi Hospital	11.569272, 77.901937	30 km/h	Medium
2	hospital	Sugumar Hospital	11.7396031, 78.0426955	30 km/h	Medium
3	clinic	R.R. Nursing Home	11.7394129, 78.0439861	30 km/h	Medium
4	hospital	Sri Hari Hospital	11.7396726, 78.0440701	30 km/h	Medium
5	hospital	Omalur Government Hospital	11.74161, 78.044296	30 km/h	Medium
6	hospital	Sri Sugam Hospital	11.7429821, 78.0461039	30 km/h	Medium
7	hospital	Saravana Hospital	11.7409303, 78.0458308	30 km/h	Medium
8	clinic	Saratha Clinic	11.7422818, 78.0458175	30 km/h	Medium
9	hospital	Arun Hospital	11.7429892, 78.0466881	30 km/h	Medium
11	clinic	Sri Meenakshi Nursing Home, Salem	11.7449162, 78.0474637	30 km/h	Medium
12	clinic	Divyam Child Care Clinic	11.7447135, 78.0473858	30 km/h	Medium
13	clinic	Momota Clinic	11.744982, 78.047397	30 km/h	Medium
14	clinic	Gomathi Clinic	11.7445293, 78.0470401	30 km/h	Medium
15	hospital	Vinayaka Child Hospital	11.745812, 78.0480225	30 km/h	Medium
16	hospital	Dr. G.R.Thilakam Hospital	11.746097, 78.04769	30 km/h	Medium
17	hospital	Thulasi Hospital	11.7461516, 78.0469984	30 km/h	Medium
18	hospital	Amirthalingam Hospital	11.7456708, 78.0470548	30 km/h	Medium
19	hospital	Children Hospital	11.7458377, 78.0480205	30 km/h	Medium
20	clinic	Shri Ramana Poly Clinic	11.7477122, 78.0479293	30 km/h	Medium
21	hospital	Dr. Nagaraj Hospital	11.7464035, 78.0472401	30 km/h	Medium
22	clinic	Manimuthu Skin Clinic	11.7521189, 78.0484058	30 km/h	Medium
23	hospital	St. Joseph's Speciality Hospital	11.761611, 78.0522344	30 km/h	Medium
25	hospital	Anugraha Multi Speciality Hospital	11.773549, 78.0546408	30 km/h	Medium

	type	name	coordinates	speed_limit	risk_level
27	hospital	L.R. Hospital	11.8703354, 78.0844554	30 km/h	Medium
28	clinic	JSA Clinic	11.8700619, 78.0844982	30 km/h	Medium
29	hospital	Thoppur,Government Hospital	11.9398824, 78.0551658	30 km/h	Medium

Crowded Spots

	type	name	coordinates	speed_limit	risk_level
0	school	KRP Matric. Hr. Sec School	11.4546193, 77.8142445	30 km/h	Medium
10	school	Fathima Hr Sec School	11.7410508, 78.0479272	30 km/h	Medium
24	school	John Britto Matriculation Higher Secondary School	11.7676907, 78.0513092	30 km/h	Medium
26	college	Dhirajlal Gandhi College of Technology	11.7968014, 78.0562095	30 km/h	Medium

Route Photos of Risky Spots



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.44024, 77.87528



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.44898, 77.87410



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.45357, 77.85789



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.45352, 77.85698



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.45847, 77.85140



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.45898, 77.85135



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.46303, 77.84945



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.45543, 77.81740



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.45631, 77.81668



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.49383, 77.88442



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.56083, 77.89921



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.56180, 77.90019



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.57048, 77.90224



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.58254, 77.90656



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.64217, 77.95200



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.64712, 77.95118



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.67858, 77.96834



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.70062, 77.97961



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.70218, 77.97866



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.71917, 78.01123



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.72565, 78.01770



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.73769, 78.03451



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.73785, 78.03551



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.73651, 78.03871



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.74265, 78.04710



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.78005, 78.05788



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.86928, 78.08384



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.88009, 78.08164



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.91277, 78.05853



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.93746, 78.05331



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.95803, 78.04964



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.95854, 78.04990



Risk Type: Turn

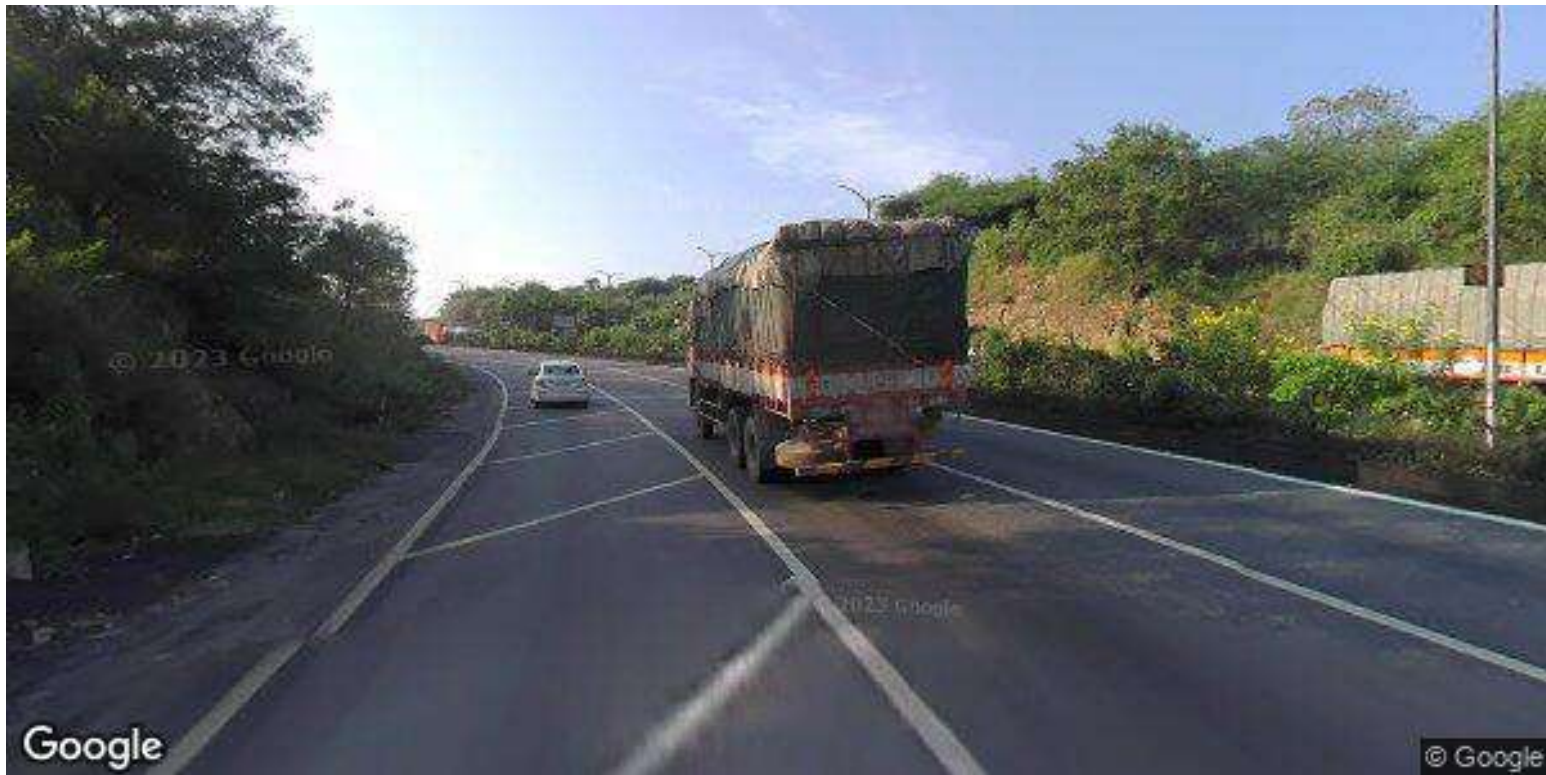
Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.96015, 78.04945



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.96370, 78.05142



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.96433, 78.05113



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.96542, 78.04938



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.96650, 78.04927



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.96804, 78.05080



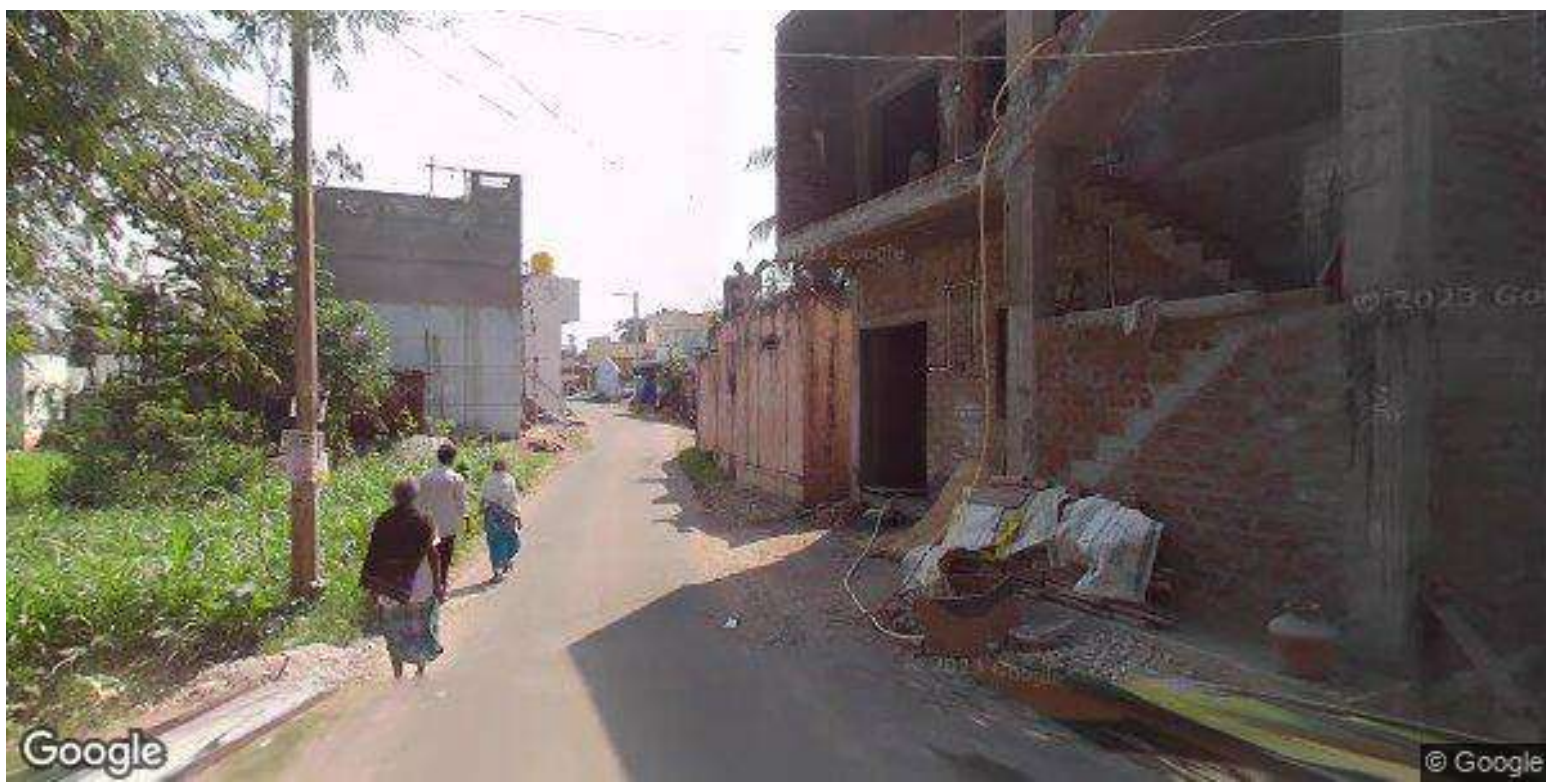
Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 12.01399, 78.08651



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 12.01914, 78.08227



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 12.02037, 78.08260



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 12.02084, 78.08193



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 12.02139, 78.08167



Risk Type: Blind Spot
Risk Level: Blind Spot
Speed Limit: 10 KM/Hr
Coordinates: 12.02306, 78.07557



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 12.02328, 78.07564



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 12.02349, 78.07557



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 12.02465, 78.07231



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 12.02566, 78.07211



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 12.02707, 78.06514



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 12.03020, 78.06332
